

Deep multi-angle research — local-dependency obtaining mechanism (ATM-064)

Revision: 1 **Last modified:** 2026-06-28T00:00:00Z **Authority:** §11.4.150 (deep multi-angle research before closure), §11.4.8, §11.4.99 (latest-source cited) **Scope:** ATM-064 — git-ignored per-host local-dependency layer + cross-platform obtaining script (scripts/obtain_local_deps.sh); first consumer = jemalloc (amber has no host jemalloc

- no sudo; mistborn has it via brew but brew is off the non-interactive SSH PATH).

Why this research (§11.4.150)

Before closing ATM-064 (and the ATM-063 jemalloc gap it subsumes), confirm — from authoritative sources, ≥ 2 distinct angles — that (i) the chosen approach (obtain libjemalloc.so.2 as a git-ignored local dep + make the binary/wrapper find it) is correct and standard, and (ii) there is no *bigger problem* we are unaware of. The hard project invariant (FACT-confirmed this cycle): **jemalloc MUST stay DYNAMIC** — scripts/tests/61_* T3 asserts objdump -p | grep NEEDED.*libjemalloc, and jemalloc is loaded via the wrapper's LD_PRELOAD. So static-linking is forbidden (would be a §11.4.1 fix-A-creates-B); the dependency must be *obtained + located*, not eliminated.

Angle 1 — Bundling a dynamic lib via rpath (DT_NEEDED resolution)

-Wl, -rpath / patchelf with \$ORIGIN is the standard way to make an executable find a bundled .so without a system install. \$ORIGIN resolves to the executable's own directory at run-time (single-

quote it so the shell/make don't expand it). This satisfies the DT_NEEDED libjemalloc resolution (test-61 invariant preserved — jemalloc stays a dynamic dep).

- patchelf --set-rpath writes **RUNPATH** by default; add **--force-rpath** to write **RPATH**. RPATH takes precedence and (unlike RUNPATH) is NOT overridable by LD_LIBRARY_PATH.
- make/autoconf mangle a literal \$ORIGIN in LDFLAGS (\$0 expansion) → set the rpath with **patchelf after the build**, or escape it — NEVER a raw -Wl,-rpath,\$ORIGIN through make.

Angle 2 — LD_PRELOAD ignores rpath/RUNPATH (THE gotcha)

man7 ld.so(8): libraries named in **LD_PRELOAD** are loaded **BEFORE the normal search begins**, and preloads do **not** benefit from RPATH/RUNPATH the way regular DT_NEEDED deps do — so a preload pathname that is not a recognised token must be **absolute** (or a \$ORIGIN/\$LIB/\$PLATFORM dynamic-string token, which LD_PRELOAD *does* understand) to be reliably found.

Consequence for us: the project preloads jemalloc, so the generated wrapper **MUST** hand LD_PRELOAD the **resolved ABSOLUTE path** to libjemalloc.so.2 (host / brew-prefix / .local-deps). rpath alone is insufficient for the preload. → The mechanism needs **BOTH**: (a) patchelf rpath on the binary for the DT_NEEDED resolution, (b) the wrapper's LD_PRELOAD/LD_LIBRARY_PATH (Linux) / DYLD_INSERT_LIBRARIES/DYLD_LIBRARY_PATH (macOS) set to the resolved absolute path. This is the §11.4.111 "resolve-by-stable-identity, not ambient PATH" discipline applied to the runtime loader.

Angle 3 — jemalloc-specific + obtain method

jemalloc upstream (INSTALL.md) supports building from source with a custom --prefix (install into a git-ignored local prefix, no sudo) and --with-jemalloc-prefix/soname-suffix for coexistence. The standard production pattern for a host that can't install it system-wide is exactly export LD_PRELOAD=/abs/path/to/libjemalloc.so.2 — i.e. obtain the .so locally + point the loader at it (precisely the operator's "git-ignored local dependency" directive). On a host with **no compiler** (amber routed to the containerised build for that reason), build/extract jemalloc inside the build container (glibc 2.35) and place the .so in the bind-mounted git-ignored prefix — amber's glibc 2.39 ≥ 2.35, so the container-built .so runs on the host.

No-bigger-problem verdict (§11.4.150(C))

No deeper defect surfaced. The only non-obvious trap — **LD_PRELOAD does not use rpath** — is real and is already handled by the design (absolute preload path + patchelf rpath for DT_NEEDED). The approach (obtain-locally + locate-via-rpath+absolute-preload, jemalloc stays dynamic) is the documented, standard one; it keeps test-61's DT_NEEDED libjemalloc invariant intact (no §11.4.1 regression) and needs no host install / no sudo on any platform.

Sources verified 2026-06-28

- ld.so(8) — Linux manual page (LD_PRELOAD load order; preload vs RPATH/RUNPATH; dynamic string tokens): <https://man7.org/linux/man-pages/man8/ld.so.8.html>
- patchelf(1) manpage (--set-rpath, --force-rpath RPATH-vs-RUNPATH): <https://manpages.debian.org/unstable/patchelf/patchelf.1.en.html>
- Creating relocatable Linux executables with RPATH \$ORIGIN (single-quote / relocatable bundling): <https://nehckl0.medium.com/creating-relocatable-linux-executables-by-setting-rpath-with-origin-45de573a2e98>
- rpath — Wikipedia (RPATH vs RUNPATH precedence; LD_LIBRARY_PATH override): <https://en.wikipedia.org/wiki/Rpath>
- jemalloc INSTALL.md (build-from-source --prefix, prefix/soname options): <https://github.com/jemalloc/jemalloc/blob/dev/INSTALL.md>
- jemalloc production LD_PRELOAD pattern (obtain + absolute preload path): <https://medium.com/@david.lowenfels/just-do-it-manually-3a7b4d151b44>